



RAFFLES INSTITUTION
Year 6 TIMED PRACTICE 2025
Higher 2

CANDIDATE
NAME

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CLASS INDEX
NUMBER

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CLASS

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PHYSICS

9749

Section B

1 July 2025

1 hour 45 min

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an 2B pencil for any diagrams or graphs.
- Write your name, index number and class in the spaces at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this section is 70.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use		
Section A	/	30
31	/	7
32	/	9
33	/	8
34	/	7
35	/	8
36	/	8
37	/	9
38	/	8
39	/	6
Deduction		
Total	/	100

Section B

Answer **all** the questions in this section in the space provided.

- 31** One end of a spring is connected to a fixed support. A mass is attached to the other end of the spring. The arrangement is shown in Fig. 31.1.

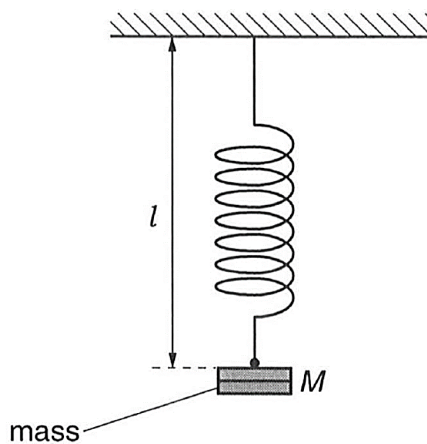


Fig. 31.1

This arrangement is used to determine the length l of the spring when mass M is attached to the spring. The procedure is repeated for different values of M . The variation of mass M with length l is shown in Fig. 31.2.

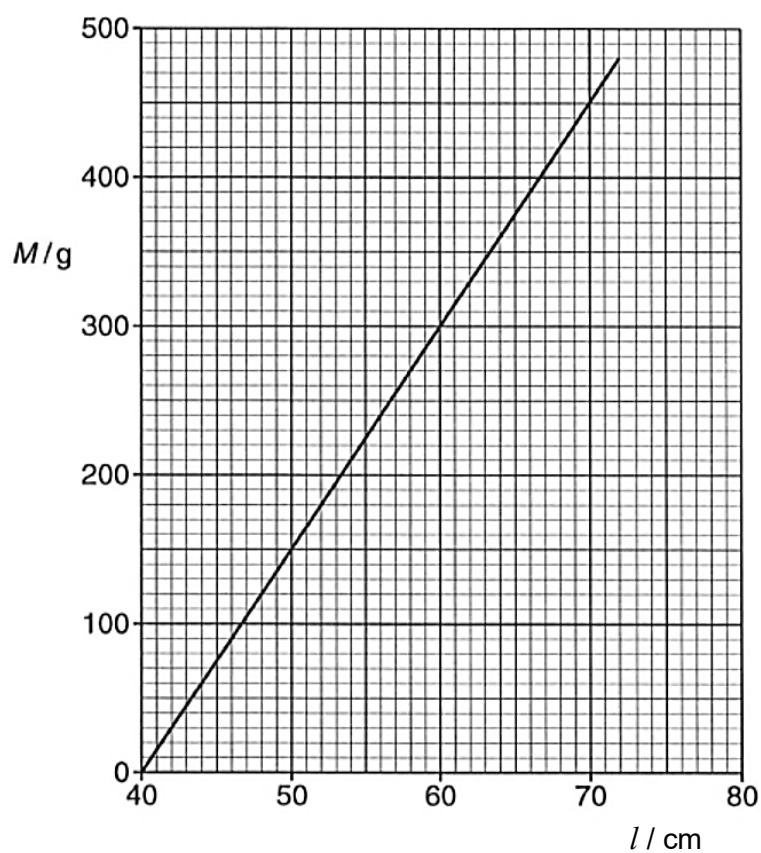


Fig. 31.2

- (a) Use Fig. 31.2 to show that the force constant k of the spring is 14.7 N m^{-1} .

[2]

- (b) A mass m of 450 g is attached to the spring and is held at rest with length l of 50.0 cm. The mass is then released, and the mass oscillates in simple harmonic motion. The angular frequency ω of the spring-mass system is given by the equation:

$$\omega = \sqrt{\frac{k}{m}}.$$

- (i) Calculate the frequency of the system.

frequency = Hz [2]

- (ii) Calculate the speed of the mass during its oscillation when the spring is extended to a length l of 80.0 cm.

speed = m s^{-1} [2]

- (c) The spring is assumed to be light. In practice, the spring will have some mass.

Assuming that the spring constant k is unchanged, suggest and explain the effect on the frequency of oscillations of having a spring with mass.

.....
 [1]

[Total: 7]

32 A loudspeaker operates at 86 Hz and produces waves of wavelength 4.0 m.

(a) Determine the speed of the wave.

speed = m s⁻¹ [2]

(b) State the speed at which regions of rarefaction and compression of the sound wave travel.

speed = m s⁻¹ [1]

(c) Another loudspeaker is now placed 20 m away to the right of the first loudspeaker. Both loudspeakers are facing each other, and a stationary wave is formed.

(i) Explain the formation of the stationary wave between the loudspeakers.

.....

 [2]

(ii) Determine the distance between any two consecutive nodes in the stationary wave formed.

distance = m [2]

(iii) With reference to the movements of molecules in a stationary sound wave, explain where the air pressure varies the least.

.....

 [2]

[Total: 9]

- 33 (a) State what is meant by *specific latent heat of vaporisation*.

.....

.....

.....

..... [2]

- (b) A student determines the specific latent heat of vaporisation of a liquid using the apparatus illustrated in Fig. 33.1.

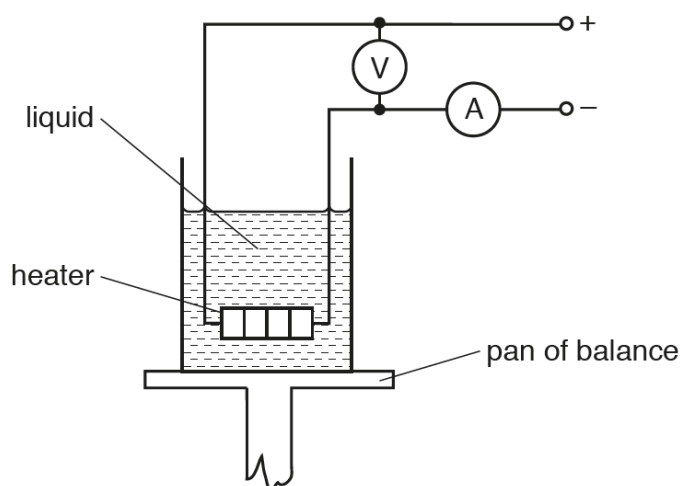


Fig. 33.1

The heater is switched on. When the liquid is boiling at a constant rate, the balance reading is noted at 2.0-min intervals.

After 12 min, the current in the heater is reduced and the balance readings are taken for a further 10 min while the liquid continues to boil.

The readings of the ammeter and of the voltmeter are given in Fig. 33.2.

	ammeter reading / A	voltmeter reading / V
from time 0 to time 12 min	1.2	230
after time 12 min	1.0	190

Fig. 33.2

The variation with time of the balance reading is shown in Fig. 33.3.

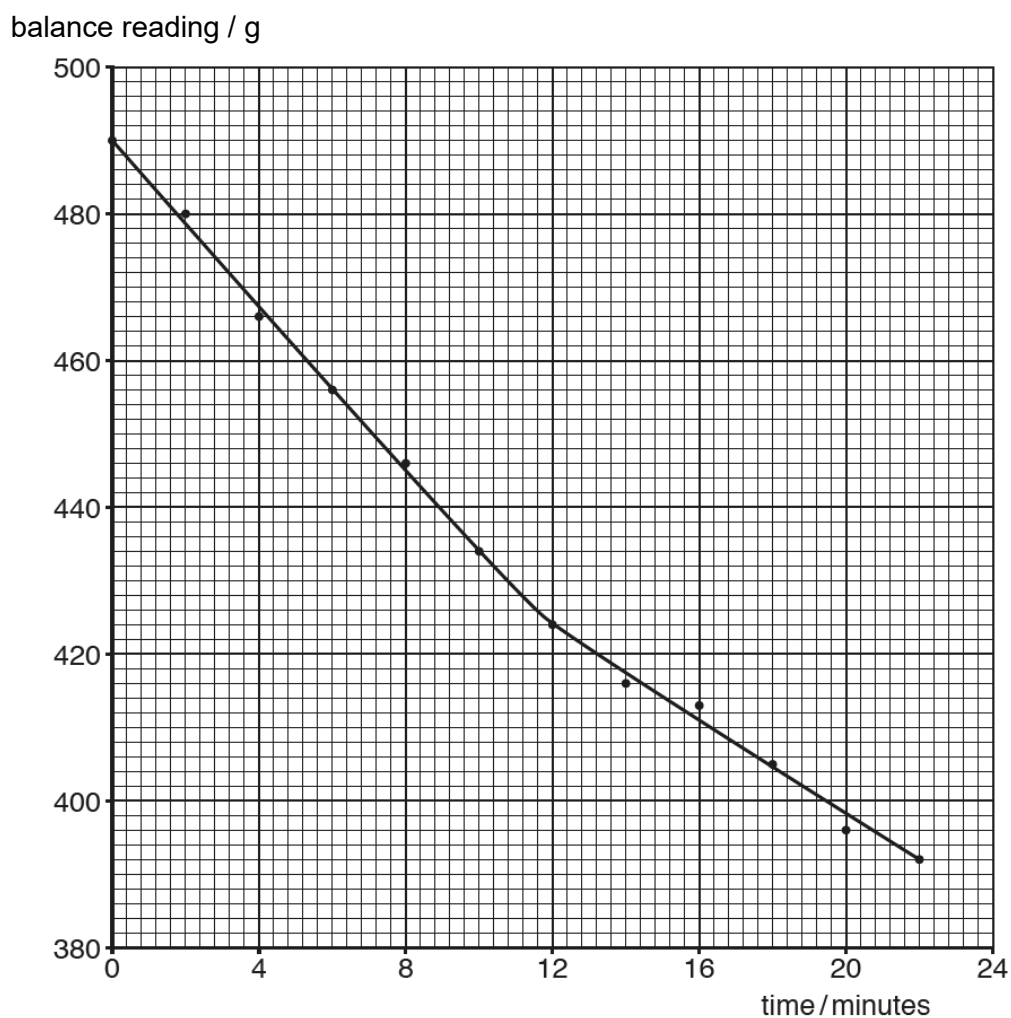


Fig. 33.3

- (i) From time 0 to time 10.0 min, the mass of liquid evaporated is 56 g.

Use Fig. 33.3 to determine the mass of liquid evaporated from time 12.0 min to time 22.0 min.

mass = g [1]

- (ii) Explain why, although the power of the heater is changed, the rate of loss of thermal energy to the surroundings may be assumed to be constant.

.....
 [1]

- (iii) Determine a value for the specific latent heat of vaporisation L of the liquid.

$$L = \dots\dots\dots \text{J g}^{-1} \quad [3]$$

- (iv) Calculate the rate at which thermal energy is transferred to the surroundings.

$$\text{rate} = \dots\dots\dots \text{W} \quad [1]$$

[Total: 8]

- 34 (a) State the relationship between electric field strength and electric potential at a point.

.....
 [1]

- (b) Two charged insulating spheres X and Y are near to each other, as shown in Fig. 34.1.



Fig. 34.1

P is a point on the line joining the centres of the spheres.

Explain why it is **not** possible for the total electric potential and the resultant electric field strength to simultaneously be zero at point P.

.....

 [2]

- (c) The magnitudes of the charges on spheres X and Y in Fig. 34.1 are Q and $2Q$ respectively. The spheres may be considered as point charges located at their centres.

Point P is a distance x from the centre of sphere X.

The electric potential at point P is zero.

- (i) Determine the distance y of point P from the centre of sphere Y in terms of x .

$y = \dots\dots\dots x$ [2]

- (ii) State an expression, in terms of Q , x and the permittivity of free space ϵ_0 , for the electric field strength E_x at P due to sphere X.

$$E_x = \dots\dots\dots [1]$$

- (iii) Determine an expression, in terms of Q , x and ϵ_0 , for the resultant electric field strength E at point P due to the two spheres.

[1]

[Total: 7]

- 35 (a) By reference to energy, explain what is meant by *electromotive force* (e.m.f.).

.....
 [1]

- (b) A battery of internal resistance r and e.m.f. E is connected to three resistors as shown in Fig. 35.1.

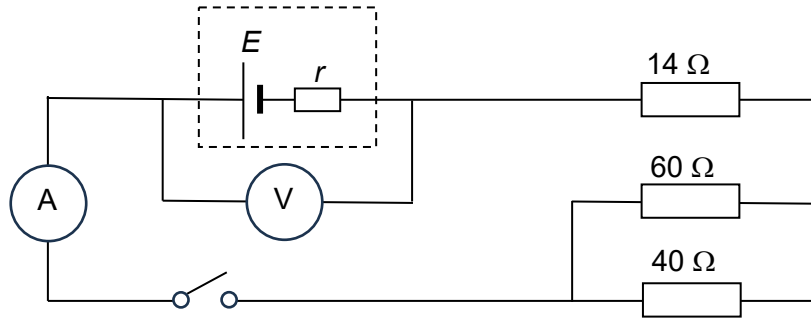


Fig. 35.1

- (i) When the switch is open, the reading on the voltmeter is 12.0 V. When the switch is closed, the reading on the voltmeter changes to 11.4 V.

Explain why the reading on the voltmeter changes when the switch is closed.

.....

 [2]

- (ii) Determine the reading on the ammeter when the switch is closed.

reading = A [3]

- (iii) Calculate r .

$r =$ Ω [2]

[Total: 8]

- 36** Fig. 36.1 shows two large parallel metal plates X and Y situated horizontally in a vacuum. Plate X is at a potential of 3000 V and plate Y at -600 V. The separation of the plates is 8.0 mm.

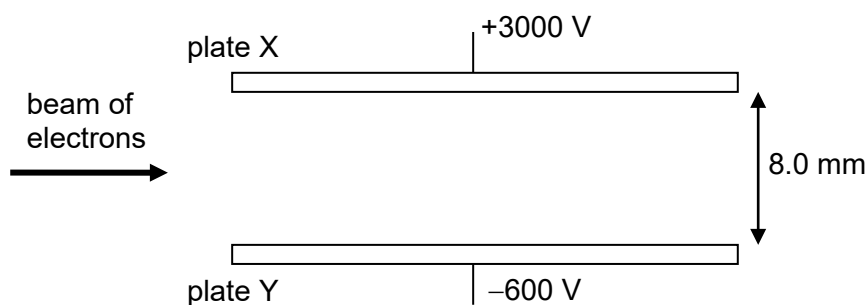


Fig. 36.1

A beam of electrons, travelling horizontally with different speeds, enters the region between the two plates.

- (a)** Determine the magnitude of the electric field strength E between the plates.

$$E = \dots\dots\dots \text{ V m}^{-1} \quad [2]$$

- (b)** A uniform magnetic field of flux density B is applied to the region between the plates, so that only electrons with a speed of $7.0 \times 10^6 \text{ m s}^{-1}$ pass through the plates undeflected.

Determine the magnitude of the magnetic flux density B .

$$B = \dots\dots\dots \text{ T} \quad [2]$$

- (c) Electrons that pass through the plates undeflected now enter a second region of uniform magnetic field of 3.5 mT and move in a circular path.
- (i) Determine the radius of the circular path.

radius = m [2]

- (ii) State and explain whether the speed of the electrons in the second region increases, decreases or remains the same if the magnetic flux density of the second region were to be increased.

.....
.....
.....
..... [2]

[Total: 8]

- 37 (a) State Faraday's law of electromagnetic induction.

[1]

- (b) A uniform magnetic field is directed normally into the plane of the paper as shown in Fig. 37.1. A coil of wire, just about to enter the magnetic field at position X, moves horizontally across the magnetic field at a uniform velocity to position Y and then position Z.

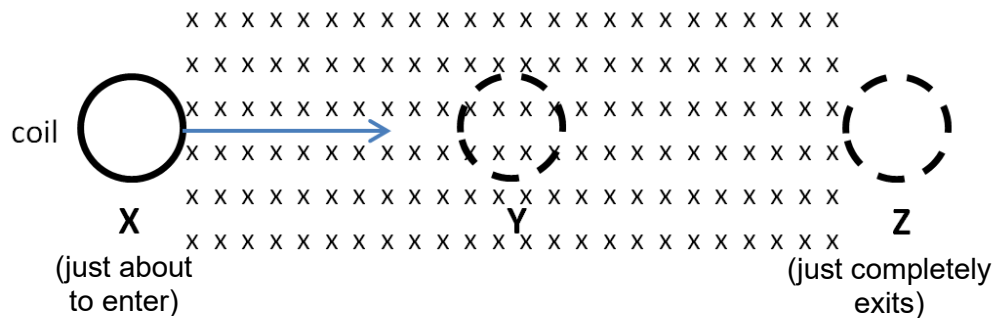


Fig. 37.1

- (i) On Fig. 37.2, sketch the variation with time of the induced e.m.f. in the coil as it moves from position X at time = 0 to position Z.

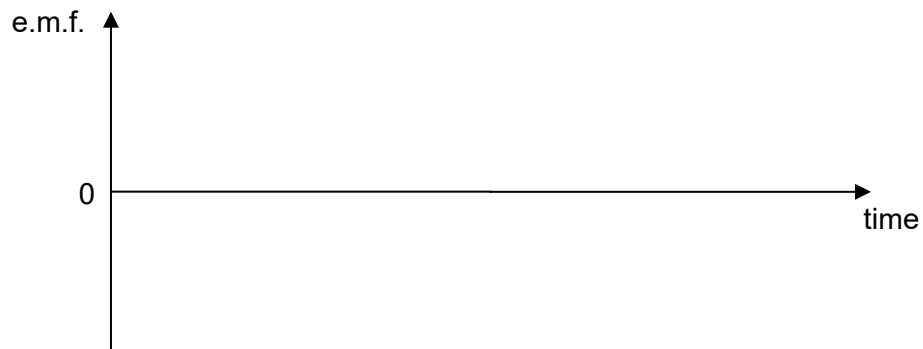


Fig. 37.2

[1]

- (ii) Explain the shape of the graph in Fig. 37.2 as the coil moves from **position X** to **position Y**.

[3]

- (iii) If the coil now moves through the field at a higher velocity, state **two** ways in which the graph in (b)(i) would change.

1

.....

2

..... [2]

- (iv) The magnetic field has a flux density of 3.0×10^{-4} T. The coil has 500 turns, a radius of 4.0 cm and a uniform velocity of 2.0 cm s^{-1} .

Calculate the magnitude of the average e.m.f. induced in the coil in the first 2.0 s of its motion.

e.m.f. = V [2]

[Total: 9]

- 38 (a) Explain what is meant by the *photoelectric effect*.

.....
 [1]

- (b) In a photoelectric emission experiment, light of wavelength 420 nm is shone on a metal surface of work function energy of 2.0 eV such that an area of 25 mm² is illuminated. A saturated photocurrent of 4.8×10^{-10} A was measured.

Determine:

- (i) the threshold frequency

threshold frequency = Hz [1]

- (ii) the stopping potential

stopping potential = V [2]

- (iii) the rate of emission of photoelectrons

rate of emission of photoelectrons = s⁻¹ [1]

- (iv) the intensity of the light source, assuming that 1 in 2500 photons succeeds in ejecting an electron from the surface.

intensity = W m^{-2} [3]

[Total: 8]

- 39** As a result of cosmic-ray bombardment of the atmosphere, the carbon dioxide in the atmosphere contains a small proportion of the unstable isotope carbon-14. Trees obtain carbon when they take in carbon dioxide from the atmosphere. When a tree is living, the concentration of carbon-14 in the tree is constant. When the tree dies, it stops taking in carbon dioxide and the carbon-14 that is present undergoes beta decay to become nitrogen-14. The half-life of carbon-14 is 5730 years.

At the time when the tree died, the activity per unit mass of atmospheric carbon due to the presence of carbon-14 averaged 0.255 Bq per gram of carbon.

- (a)** Define *half-life*.

.....
 [1]

- (b)** Determine the number of carbon-14 atoms per gram of carbon when the tree died.

number of carbon-14 atoms per gram = g^{-1} [2]

- (c)** A specimen of the tree contains 500 mg of carbon and its activity is 174 per hour.

Determine the number of years that passed since the tree died.

number of years = years [3]

[Total: 6]